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PAPER_963: Buoyancy Gravity Triadic Mode

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic_{solutions_next}.py (BuoyancyGravityTriadic) **Calculator:** BuoyancyGravityTriadicCalc (CP4 #547) **CVW:** v2.0.0 compliant

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Abstract

The Buoyancy Gravity Triadic mode evaluates $E_{\mathrm{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_t extSCmt) \cdot \exp(-\Gamma t) - \text{threshold}$. Positive E_{net} drives expansion (nebulae, HII regions); negative E_{net} drives erosion (filaments, cometary knots).

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1. Net Energy

$$E_{\mathrm{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_t extSCm \cdot t) \cdot \exp(-\Gamma \cdot t) - \text{threshold}$$

2. Regime Classification

E_{net}	Regime	Astrophysical Example
> 0.1	Expansion	Nebulae, HII regions
< -0.1	Erosion	Filaments, pillars
[-0.1, 0.1]	Neutral	Transition zones

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_961 — Compressed Gravity Triadic
3. PAPER_962 — Resonant Gravity Triadic
4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch of same triadic
PAPER_962	Resonant branch

Related Paper	Relationship
PAPER_966	Unified solver combining all three
PAPER_954	$E(t)$ sign-flip from buoyancy dynamics

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda\left(\phi^2 - v_{\text{SCm}}^2\right)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\, v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy coupling
E_{net} sign	—	$+$: expansion, $-$: erosion	Phase indicator

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
E_{net} sign-flip	Expansion $\leftrightarrow$ erosion phase transition	Derived
F_{UBi} buoyancy force	Acts outward against g_{comp}	Validated

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Buoyancy Gravity (F_{UBi} Net Energy Balance)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{buoy}} = F_{UBi}(r) \cdot r - \int_0^r g_{\text{comp}}(r')\, dr'$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{E_{\text{net}}(r) = F_{UBi}(r) \cdot r - \int_0^r g_{\text{comp}}(r')\, dr', \quad \text{sign}(E_{\text{net}}) : +\text{expand}, -\text{erode}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\to$ buoyancy force F_{UBi} $\to$ net energy balance $\to$ expansion/erosion phase $\to$ triadic branch 3/3

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

F_{UBi} profile traces VDS outward pressure gradient.

§B.2 DVP

Buoyancy vortex dipole: inward gravity vs. outward F_{UBi}.

§B.3 BSH

E_{net} saturates at stellar surface (BSH envelope crossing).

§B.4 Production-Scale Consistency

Metric	Value	Status
β_i	0.603	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

6 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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